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GROUP Assignment Cover Sheet

Course Code:	RSM 8502	**Student Numbers**: *Please list all student numbers included in this group assignment.*	1002590341 1009687572 1004820759 1004797428 1008249637
Course Title:	Data-Based Management Decisions		
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Assignment Title:	RSM 8502 Group Assignment		
Date:	September 12th, 2022		

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- All members have contributed substantially and proportionally to this assignment.
- All members have sufficient familiarity with the entire contents to be able to sign off on this work as original.
- This is the final version of the assignment and not a draft.
- We have followed any specific formatting requirements set by the instructor.
- We are submitting this work for the correct course, via the specified platform/method (e.g., Quercus).
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Complete Course Code (including Section) – GROUP – **Assignment Title**

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Introduction:

DT Bank, one of the leading banks in Canada, has been running a campaign to increase its revenue from selling fixed-income securities. The main purpose of the campaign was to collect relevant client data associated with the outbound sales calls that financial advisors made to reach out customers. In this report, we focused on assessing whether the campaign was successful in increasing sales and developing a prediction model using the campaign data as the recommender system to identify potential customers. Moreover, we aimed to make recommendations on whether DT bank should purchase the additional socio-economic data which was hoped to further increase sales.

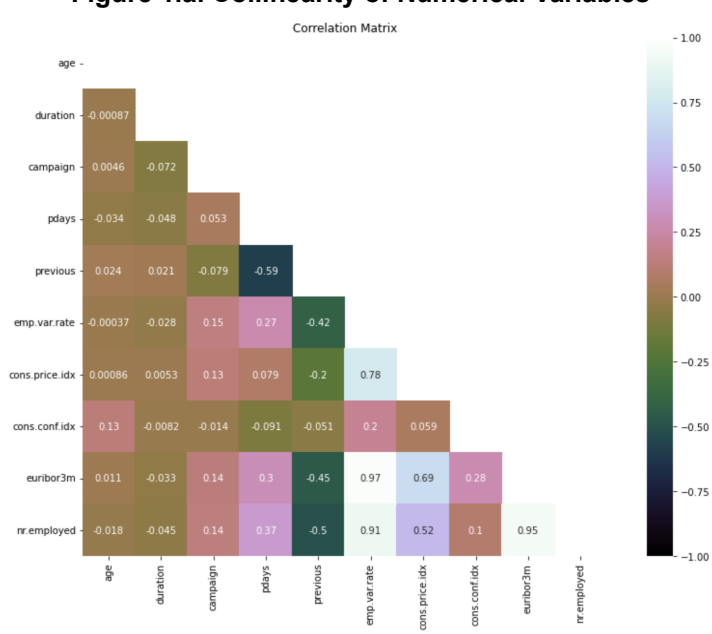
The remaining parts of the report were organized in the following way: Section 1 addressed exploratory data analysis with discussion on collinearity, missing values, preliminary cross-tabulation analysis among categorical variables from group 1 to group 3, and conclusions on whether the campaign was successful. Section 2 focused on constructing a classification model as a recommender system to predict the customer's purchase behavior using selected variables from group 1 to group 3. Section 3 discussed the classification models on two separate datasets based on scenarios that included or excluded the additional socio-economic variables. In the end, after evaluating outputs and analyzing their implications, we have made detailed recommendations for DT bank.

Section 1: Exploratory Data Analysis

1.a. Collinearity

After loading the data, we have constructed the correlation matrix and looked for collinearity between variables. Moreover, for better demonstration, we have plotted the correlation matrix in heat maps to visualize the relationship as shown in *figure 1.a.* below.

Figure 1.a. Collinearity of Numerical Variables



1.a. Collinearity (continued)

Strong linear relationships are defined as having a correlation coefficient greater than 0.7 between two quantitative variables. In the heat map shown above, all variables shown were numerical and we can see that 4 pairs of variables exhibit strong linear correlations and 1 pair of variables is very close to having a strong linear correlation. The pairs are:

- Employment variation rate vs. Number of employees, 0.91
- Employment variation rate vs. Euribor 3-month rate, 0.97
- Employment variation rate vs. Consumer price index, 0.78
- Consumer price index vs. Euribor 3-month rate, 0.69
- Euribor 3-month rate vs. Number of employees, 0.95

1.b. Missing value analysis

To find missing values in our dataset, we first started with searching for “n/a” or “null” values in the columns. After searching, we found no counts on variables with either “n/a” or “null” values. Therefore, we concluded that there were no missing values for the numerical variables but further examinations were needed for categorical variables as we could not rule out the possibility that certain strings were used to indicate the missingness. After examining the distributions of values for the categorical variables, we found the string name “unknown”, which indicated missing values, presented in “job”, “marital”, “education”, “default”, “housing”, and “loan” as shown in the figure 1.b (*Figure 1.b.*). To deal with missing values, several approaches are applicable for different scenarios. In our case, several options were raised to our consideration, including deletion of missing data, assigning indicator variables to represent missingness, mean imputation, and “proportion replication”.

Generally speaking, it is unlikely to have the “missing at random” situation in a dataset and hence simply deleting missing values would lead to bias in the dataset and jeopardize the precision of the model. Hence, imputations were more suitable in our analysis to reduce biases in the dataset. Assigning an indicator variable to the missing value is reasonable when the missing value itself is informative in the model, in the case of variable “default”, the number of missing values were considerably large compared with other variables and we agreed that “unknown” itself can be categorized to represent a group of customers. In addition, since “unknown” responses were assigned categorical values in the “default” variable, we chose to keep them without further transformation. As mentioned above, missing values only existed in the categorical variables so mean imputation was not applicable in our analysis. Lastly, we arrived at another method which we thought would be the most reasonable to use in our EDA, the “proportion replication” method.

1.b. Missing value analysis (continued)

We first looked at the distribution excluding the missing values, and then applied the same distribution to the missing data and we ended up with more counts on other categorical values while maintaining the similar distribution as before, as shown in the figure 1.b.1 (*Figure 1.b.1.*). We did the transformation for “job”, “marital”, “education”, “housing”, and “loan”. The number of missing values we found in each variable were relatively small in most cases and we found no reason to assume they would be skewed in a particular way, thus we assumed that they follow the same distribution as other categorical values.

1.c Cross-tabulation analysis

A frequently used method to quantitatively analyze the relationship between multiple categorical variables is cross-tabulation. As shown in *Table 1.c.* in the Appendix, a contingency table was constructed for each categorical variable with respect to the response y. Each cell represents the count of cases that belong to the pre-defined category, and we performed Chi-Square hypothesis tests to determine whether the predictors were significantly associated with the target variable y.

For example, the cross-tabulation table for the “contact” variable gave a simple glimpse that DT Bank has achieved approximately five times more sales from clients who were contacted through cellular communications (3853 cases) than those who were contacted by telephone (787 cases).

Yet, the fact that the majority of clients did not purchase the fixed-term deposits was apparent as Table 1.c presents due to the huge difference of counts in clients who said “yes” and “no” to the campaign, regardless of the communication style. Underneath each table, the statistics for the hypothesis test was provided: Since large Chi-Square value and small p-value (≤ 0.05) suggest high dependency between two variables, we could conclude that communication method was significantly correlated with the success of DT bank’s campaign. We can also gain some insights on other variables that should catch our attention: For instance, most of the phone calls were made from May to August and corresponded to a higher number of successful sales, phone calls that were made on Wednesday and Thursday were more likely to lead to sales.

From the cross-tabulation analysis, we identified that job type, marital status, education level, whether having default history, contact method, month, day of week, and previous campaign outcome all had strong association with the target variable, whereas housing loans and personal loans did not show significant influence on y. While it is not uncommon to relate the clients’ job type and education level to their capability to buy the fixed-term deposit, it turns out that month and day of week of the phone calls have a more manifest influence on the success rate. These ideas will be considered in detail when we fit predictive models for the DT Bank in the following sections.

1.d Evaluation of results

Based on our current explanatory analysis on the relationship between independent variables and the target y, we believe that the variables in group 2 would be most useful in explaining y, namely contact, day of week, month, and duration. Through the cross-tabulation analysis in the previous part, we first eliminated group 1 from consideration since two of the variables in the group were not significant in explaining y, whereas in group 2 and 3, all four variables had statistically significant association with the response variable. It leaves us a choice between the remaining two groups, and a further multivariate analysis was conducted to explore the relationship of multiple factors on y.

As shown in the following *Table 1.d*, we investigated the combined effect of group 2 or 3 on the target y. The result yielded a clear view that the variables in group 2 remarkably outperform group 3, despite the fact that both of them have low explanatory powers. The value of R-Square for group 2 was 0.281, which means it could only explain 28.1% of variation in y. However, it still outperformed group 3, which had a R-Square value as low as 0.096. In addition, we have calculated the Area Under Curve (AUC) for each group to explore the classification ability for both models. Based on the result, group 2 turned out to have higher accuracy with AUC equaled to 0.883 while group 3 had a notably lower probability (AUC = 0.644) to correctly classify whether a client would buy the fixed-term deposit. Thus, in both explanatory and predictive aspects of the model, group 2 seemed to be a better choice for analyzing client behaviours in response to the campaign.

Table 1.d. Comparisons of Performance of Variables in Group 1-3

	R Squared	AUC
Group 1	0.0457	0.6515
Group 2	0.2814	0.8828
Group 3	0.0962	0.6436

By investigating the proportions of sales generated from the campaign, we saw that even though the proportion of sales only accounts for 12.7% of total phone calls, it yielded a quite successful outcome compared to the average success rate of approximately 2% from the *64 Cold Calling Statistics You Must Learn* study by Allan Jay. For the current campaign, 4640 of the clients purchased the fixed-term deposits, which was about 4 times more than the customers who bought the product from the previous campaign (1373 cases). This could be a remarkable improvement from the last campaign with a significant increase in the number of successful sales and the expanding scale of targeted customers. Despite the possible cost of marketing and labor to increase the sales for the fixed-term deposit, the current campaign seemed successful in a context where it was compared to the average industry benchmark.

However, without knowing what target benchmark successful rate DT Bank was looking for, simply looking at the proportions was insufficient to conclude the performance of this campaign. Therefore, it was important for us to develop a recommender system using data modelling approaches to effectively predict customer behaviours and specify targeting segmentations to improve performance of the DT Bank campaign.

Section 2: Predictive Modelling

2.a. Building a predictive model

The problem mainly focuses on predicting the success or failure of calling a customer given the parameters of interest. In order to use predictive modelling, logistic regression should be the apt model for this case to predict the success of contacting a customer. Given the fact that the target variable is a binary variable, we are using a binary logistic classification model to predict the outcome whether the sale has been made. We consider logistic model for some of the following rationales:

- It takes into account the input features contributing for the success of the calling and predicts the nearest possible outcome
- The sigmoid function of the logistic is the function which calculates the probabilities of the value between 0 and 1, and hence it is used in models where the output is a probability
- The logistic model has a threshold which can be used to classify the outcome based on it. Also, the classification rate of the model can be used to understand the effective classification point to be considered

We have built the classification model based on several sets of variables and identified that most of the effects of variables utilized in the process are significant except for loan and housing. The variables used in the modelling exercise are ("month, contract, poutcome*month, default, job, campaign, dummy_pdays, day_of_week, poutcome, marital, previous, age, education, loan, housing). Of all these variables there are some interaction variables and transformation variables considered for analysis. Though we mention that the group 2 variables are better in predicting y, it seems like after analysing the coefficients of the modelling variables most of the group 3 variables are contirbuting higher to the prediction of y.

The test-train split has been considered before modelling the data and the data has been split to 20% for test and 80% for train. The training set was used to fit our predictive model and for the selection of the optimal decision threshold, and the trained model on the 80% data is utilized for predicting the outcome on the test and we further compared the predicted and the test values to arrive at conclusions. The validation set was used to provide the out-of-sample performance evaluation. We sufficiently prevent our models from overfitting and better evaluate our models by doing so.

The variable dummy_pdays is created with the idea of applying transformation to the pdays. The value for the pday indicates 'the days passed after the client was last contacted'. We have values '999' in the column which indicates the customer was not contacted previously. We find that the variable is fit to be transformed into a category and used in the modelling phase. For the entry which is '999', we substituted '0' and for the rest we used '1' as the entry and called it as dummy_pdays. The effect of doing this improves the coefficient of the variable from -0.0025 to -0.614 thus increasing the explainability.

2.a. Building a predictive model (continued)

For the default variable there are three categories ('yes', 'no' and 'unknown') of which we did not remove the third category and retained the unknown category. This seems to be a valid entry and we utilized this variable in the modelling. One interaction we found worth using in the model is the outcome and month interaction which seems significant in terms of the contribution to the y variable prediction. Hence we considered the case. Below is the table of variables used in the modelling.

2.b. Effect of duration in prediction

We have conducted modelling using duration and without duration. Duration can not be used to help the marketer choose which consumers to call in advance, and has less relevance in terms of the model variables used for analysis. It only makes sense when the campaign has happened. Hence, we excluded the usage of duration variables in modelling.

2.c. Output analysis of the model

The overall model has a good classification rate and the output of the model seems to be almost similar for the train and test sets. From the results of table 2.c (Appendix)., it can be seen that the LLR p-value (0.000) and the Pseudo R-squ is 0.1246. The p value measures the statistical significance, which is <0.05, hence proves the model is significant. From the same table, when looking at the coefficients, we notice that the previous(cf = 0.389, pvalue=0.00), poutcome(cf = 0.1954, pvalue = 0.042) and marital (cf = 0.2429, pvalue = 0.00) are contributing to good prediction of the y variable.

The below table shows the evaluation metrics for the train and test split of the data and as the data has a greater predictor variable contributing to "no"(36548) than "yes"(4640), the specificity of the variable is high compared to the sensitivity of the data.

2.d. Predictive performance evaluation

By evaluating the logistic regression model, there are two indicators that are important for the measure. One is p value, and the other is Pseudo R-squ. Noticing the value of pseudo R-square (0.1246) is not very high, the p value is the most important one to explain whether the model is significant or not, and as stated in the output analysis, it is significant. In addition, to assess the performance deployment, we identified the optimal threshold of the model during training and utilized the same for evaluation of the test data. Figure 2.d.i and 2.d.ii (Appendix) shows a confusion matrix that depicts the performance capability of the models for train and test data. Based on the confusion matrix tables, we calculated and compared the sensitivity and specificity merics, and found that the model performs relatively well on the test set as it does on the training set.

In addition,as we see from table 2.a below, the train data to have a slightly greater accuracy, we notice that the test accuracy is equivalently good to the train. The thresholds for the model can be found to be 55% as we used the corresponding highest classification rate as a measure to consider the threshold. The ROC curve (Figure 2.d) for the first model depicts

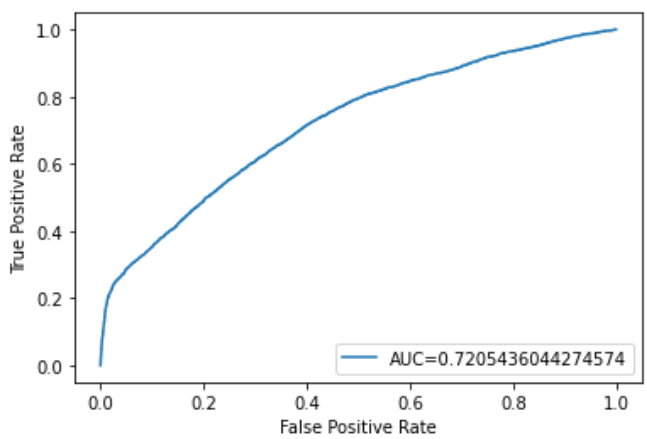
that the Area under the curve (AUC) was 0.721, indicating the model performed well at identifying the classes and can achieve a high specificity score (rate of identifying 'no' as 'no' without needing to trade-off much of its specificity (rate of identifying 'yes' as 'yes'), known the the data has a greater predictor variable contributing to "no"(36548) than "yes"(4640),the model has a lot better specificity than sensitivity.

Also in Table 2.d.3.(Appendix), we have presented different classification frequencies for different thresholds.

Table 2.a Test-Train comparison table for model 1

Group (1-3)	Accuracy	Sensitivity	Specificity	Threshold
Train	89.72%	18.292%	99.871%	55%
Test	89.47%	15.840%	99.823%	

Figure 2.d. ROC curve for test data Group (1-3)



Section 3: Sensitivity Analysis

In this section, we evaluated the predictive ability of socio-economic data on DT bank's clients' purchasing behavior to better understand its contribution to the conversion of the outbound sales calls. This would help DT bank decide whether it was worth purchasing at a cost of $5000 dollars. To achieve the above goal, sensitivity analysis was conducted on the socio-economic variables in group 4 through model comparisons and cross-tabulation analysis.

3.a. Predictive model with Socio-economic Data

First, we started with clarifying the definitions for all 5 variables in group 4. Employment variation rate represented the changing rate of employment within a quarter, which had a range from -3.4 to 1.4 and most of the data was above 0. The consumer price index measured

the average change in the price of a market basket of consumer goods and services over a certain period of time, which varied from 92.2 to 94.77 in the given dataset.

The consumer confidence index indicated consumers' sentiment about their expected financial situation and the general economic situation, which might influence a household's future consumptions and savings and in this case, had a range from -50.8 to -26.9. Euribor 3-month rate was the benchmark rate which comprised the average of the overnight interest rate from a group of leading European banks, which had a range from 0.634 to 5.045. The number of employees, as its literal meaning, showed the number of employed workers by quarter with a range from 4964 to 5228.

From the correlation matrix heat map as shown in figure 1.a, we found that the employment variation rate was highly correlated with Euribor 3-month rate, number of employees, and consumer price index with correlation coefficient of 0.97, 0.91 and 0.78 respectively. This indicated that all these four variables might reflect very similar information in a predictive model. On the other hand, we did not find the consumer confidence index had significant correlation with other variables. Thus, we decided to only add employment variation rate and consumer confidence index to the original logistics regression model for the purpose of evaluating the effect of socio-economic data from group 4.

3.b. Performance evaluation for the predictive model

Even though we found that the pseudo R-squared in the new model had increased 39.5% from 0.1246 to 0.1730, it was still considerably low in terms of the explanatory power on the variation of the outcome variable. The new logistic regression model had the best accuracy rate of 89.41% with the cutoff of 0.53 in the test dataset. Compared to the model from section 2 with only variables from the first three groups, there was a slight decrease around 0.06% on the accuracy rate despite the AUC having an increase from 0.72 to 0.77 (*Table 3.b.*). The table below (*Table 3.a*) depicts the test and train split of the model 2 using group 4 variables in addition to group 3 variables.

Table 3.a. Test-Train comparison table for model 20

Group (1-4)	Accuracy	Sensitivity	Specificity	Threshold
Train	89.78%	17.968%	98.905%	53%
Test	89.41%	15.301%	99.823%	

To profit from investing $5000 dollars on the additional socio-economic data, and given that DT bank was willing to pay $1000 dollars for each percentage increase in model performance, it meant that DT bank would need to secure at least 5% improvement on the model performance. However, considering accuracy rate as the most important improvement criteria, the new model was not performing 5% better. As a result, introducing additional

socio-economic data in this case could not generate the required return. Therefore, it was reasonable for DT bank to not spend $5000 on the additional socio-economic data.

3.c. Cross-tabulation analysis on Socio-economic Data

Moreover, we decided to perform cross-tabulation analysis on all five variables in order to figure out the dependency between socio-economic data and the success of sales calls. However, the numerical nature of these variables required pre-processing before conducting cross-tabulation analysis. The conversion of numeric variables to categorical variables had been performed through dividing the numerical values into proper sets of category buckets. As shown in *table 3.b*, all results from the five cross-tabulation analyses came with a very high Pearson's chi-square statistic with p-value close to 0, which indicated that all five variables in group 4 have had a statistically significant dependency with outcome Y. However, it is known that the chi-square is very sensitive to sample size. Even trivial relationships could appear to be statistically significant with a large sample that is bigger than 500. The result we obtained for the five variables might not be meaningful and reliable. Also, as shown in the previous model comparisons, the predictive ability of the socio-economic data is very limited.

Limitations and Recommendation

Based on the results we obtained from using two different sets of models, we learned from the output variable that there were 36,548 responses on "No" and 4,640 responses on "Yes". The number of "Yes" was comparatively lower than the number of "No", which indicated that the dataset had an imbalanced distribution of the outcome variable and it could result in low sensitivity and high specificity. Sensitivity means the rate of correctly predicting the actual "yes scenario" while specificity means correctly predicting the "No scenarios". In our case, DT bank was in need of a predictive model to capture customer's inclination to buy the fixed-income security (the "yes" scenario) so we would be looking for sensitivity instead of specificity. However, as shown above, it would be a challenge for the current model to be informative because of the low sensitivity resulting from the imbalance distribution of the outcome variable. Therefore, we propose two forward approaches to address the issue of imbalance in the dataset. SMOTE (synthetic minority oversampling technique) and ROSE (random oversampling examples) can be applied in future analyses since they all specialize in solving the undersampling issue by synthesizing examples from the minority classes to balance the distribution.

Secondly, we impute the missing values with "proportion replication" combined with assigning categorical indicators to "unknown". Although our method may be applicable for this report where missing values take a relatively small portion of the entire population, more sophisticated algorithms are worth exploring to further reduce biases in the dataset. For example, methods such as multiclass logistic regression and KNN (k-nearest neighbors algorithm) may provide stronger power in reducing biases, but they would require significant efforts in setting up the initial model and it was beyond the scope of our focus for this report.

Conclusion

We began with exploratory data analysis on correlation analysis on numeric variables and cross-tabulation analysis on categorical variables to discover the relationship among variables. We chose to deal with missing values using “proportion replication” and retained the categorical value “unknown” for the variable “default”. Going forward, we discovered that variables in group 2 performed better in predicting the outcome variable y as the initial starting point for the following analysis. Then we concluded that the DT Bank had a higher success rate of sales conversion than the benchmark success rate on the cold-call statistics we found in the industry, which suggested that the campaign was successful from the demonstration of higher-than-industry performances. However, a recommender system using predictive models was crucial to secure customers and generate long-term sustainable sales performance.

Subsequently, the binary logistic regression was chosen to predict the likelihood of success of the sales calls as it contributes to having a threshold given our binary nature of the outcome variable and the tendency of the sigmoid function in prediction of the outcome. Some transformation and interaction on the variables were added to the model in order to improve the model's predictive performance. Moreover, we found that the socio-economic data was not helpful in increasing the predictive performance of the predictive model and thus was unable to help DT bank to yield the required return. As a result, we concluded that DT bank should not invest on the additional data. In the end, we addressed some of the limitations of our report and proposed recommendations on future modifications of the analysis.

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2. N.V. Chawla, K.W.Bowyer, L.O. Hall, W.P. Kegelmeyer. n.d. “SMOTE: Synthetic Minority Over-Sampling Technique.” *Journal of Artificial Intelligence Research* (2022): 321–57.https://doi.org/10.1613/jair.953.
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Appendix

Figure 1.b. Summary of Missing Values

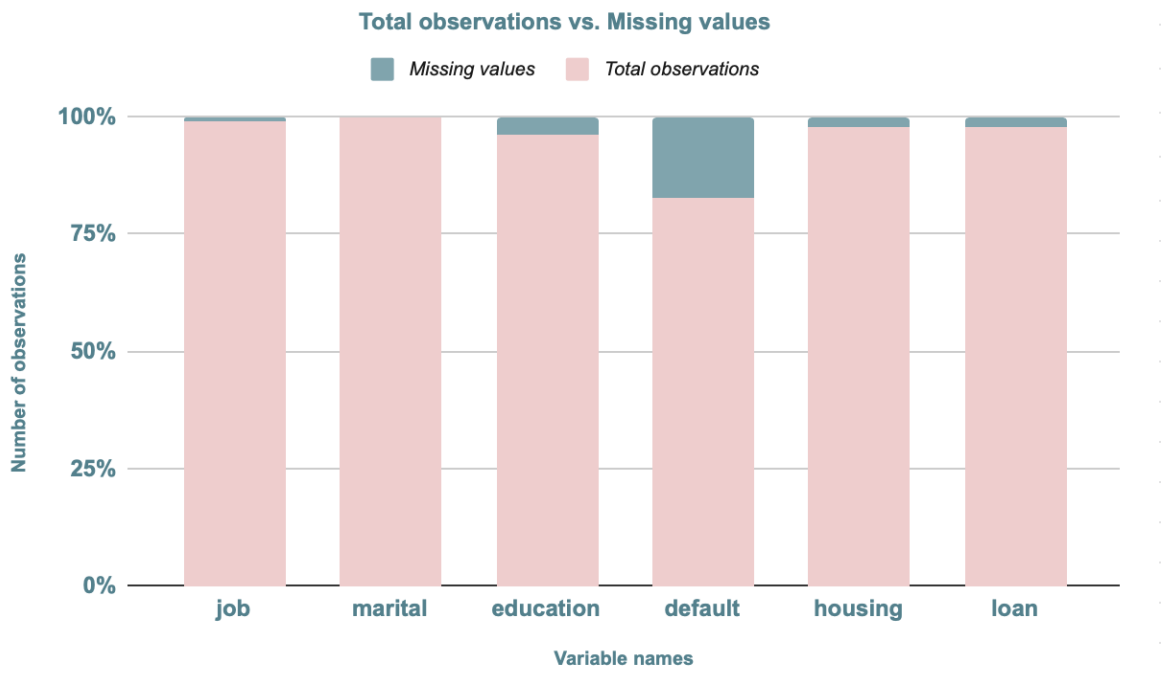

Figure 1.b.1 Illustration of Missing Value Transformation

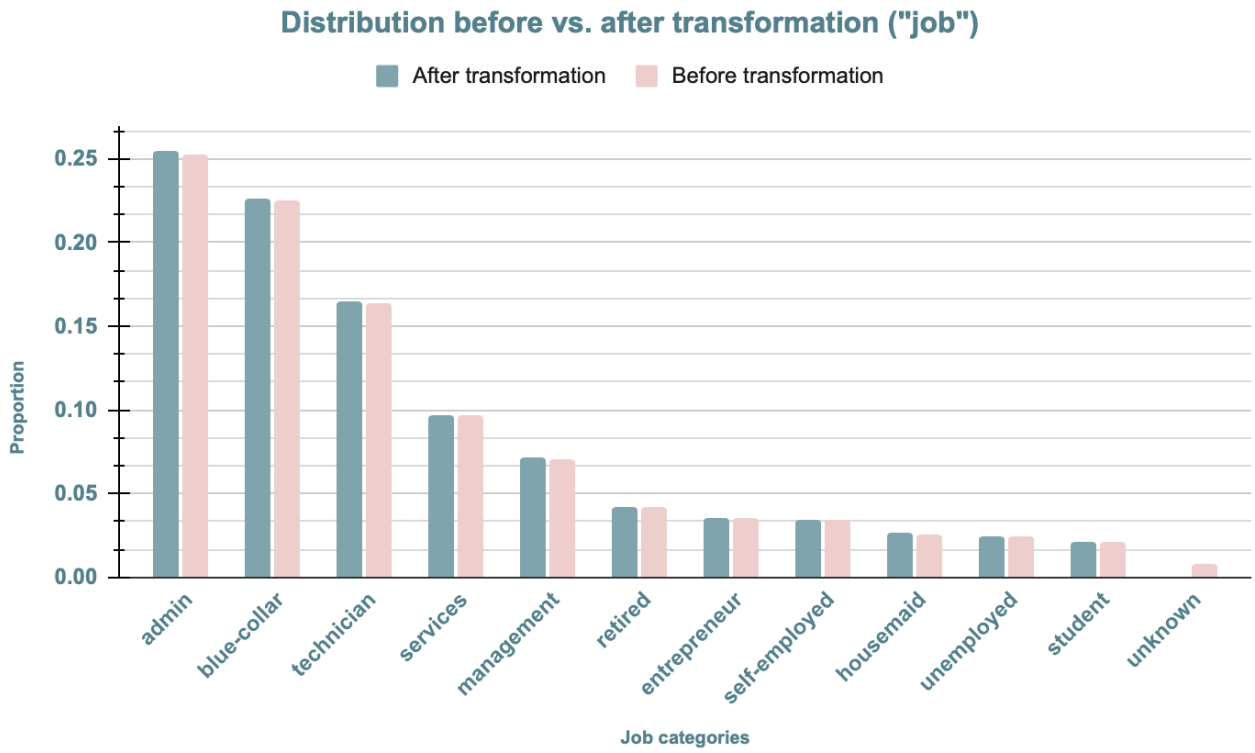

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Table 1.c. Cross Tabulation of Categorical Variables

Group 1 - Demographic Attributes

Job - Imputed

y	admin	blue-collar	entrepreneur	housemaid
no	9140	8693	1341	961
yes	1364	646	125	108
	management	retired	self-employed	services
no	2615	1300	1281	3680
yes	328	435	150	328
	student	technician	unemployed	
no	605	6055	877	
yes	276	735	145	
Chi-Square	**954.816**			
P-value	**7.992E-188**			

Marital - Imputed

y	divorced	married	single
no	4141	22434	9973
yes	477	2540	1623
Chi-Square	**120.502**		
P-value	**1.278E-23**		

Education - Imputed

y	basic.4y	basic.6y	basic.9y	high.school
no	3878	2192	5808	8827
yes	454	202	514	1087
	illiterate	professional.course	university.degree	
no	14	4832	10997	
yes	4	632	1747	
Chi-Square	**163.625**			
P-value	**1.323E-27**			

Default

y	no	unknown	yes
no	28391	8154	3
yes	4197	443	0
Chi-Square	**406.578**		
P-value	**1.077E-84**		

Housing - Imputed

y	no	yes
no	17007	19541
yes	2074	2566
Chi-Square	**5.576**	
P-value	**0.233**	

Loan - Imputed

y	no	yes
no	30833	5715
yes	3944	696
Chi-Square	**1.271**	
P-value	**0.866**	

Group 2 - Last Contact

Contact

y	cellular	telephone
no	22291	14257
yes	3853	787
Chi-Squared	**863.269**	
P-value	**1.512E-185**	

Month

y	mar	apr	may	jun	jul
no	270	2093	12883	4759	6525
yes	276	539	886	559	649
	aug	sep	oct	nov	dec
no	5523	314	403	3685	93
yes	655	256	315	416	89
Chi-Squared	**3101.149**				
P-value	**0.0**				

Day of Week

y	mon	tue	wed	thu	fri
no	7667	7137	7185	7578	6981
yes	847	953	949	1045	846
Chi-Squared	**26.145**				
P-value	**0.004**				

Group 3 - Historical Campaign

Poutcome

y	failure	nonexistent	success
no	3647	32422	479
yes	605	3141	894
Chi-Squared	**4230.524**		
P-value	**0.0**		

Table 2.c Coefficient table for model 1

MNLogit Regression Results

Dep. Variable: y	**No. Observations: 41188**
Model: MNLogit	**Df Residuals: 41172**
Method: MLE	**Df Model: 15**
Pseudo R-squ.: 0.1246	**Log-Likelihood: -12693.**
converged: True	**LL-Null: -14499.**
Covariance Type: nonrobust	**LLR p-value: 0.000**

Table 2.c Coefficient table for model 1 (continued)

y=1	coef	stderr	z	P>|z|	[0.025	0.975]	Explainability
Intercept	-1.1141	0.284	-3.918	0.0000	-1.671	-0.557	0.328211
age	0.0158	0.002	9.55	0.0000	0.013	0.019	1.015925
default	-0.7112	0.054	-13.104	0.0000	-0.818	-0.605	0.491055
contact	-0.8416	0.044	-19.187	0.0000	-0.928	-0.756	0.43102
day_of_week	0.0284	0.012	2.355	0.0190	0.005	0.052	1.028807
campaign	-0.087	0.009	-9.361	0.0000	-0.105	-0.069	0.916677
previous	0.389	0.054	7.188	0.0000	0.283	0.495	1.475505
poutcome	0.1954	0.096	2.037	0.0420	0.007	0.383	1.215797
job_imp	0.0091	0.005	1.947	0.0520	-6.09E-05	0.018	1.009142
marital_imp	0.2429	0.03	7.968	0.0000	0.183	0.303	1.274941
education_imp	0.0361	0.009	4.239	0.0000	0.019	0.053	1.03676
housing_imp	-0.0099	0.034	-0.295	0.7680	-0.076	0.056	0.990149
loan_imp	-0.0395	0.047	-0.85	0.3950	-0.131	0.052	0.96127
month	-0.0341	0.014	-2.448	0.0140	-0.061	-0.007	0.966475
pdays	-0.0754	0.007	-10.28	0.0000	-0.09	-0.061	0.927372
poutcome:month	0.0385	0.013	2.987	0.0030	0.013	0.064	1.039251

Figure 2.d.1 Confusion matrix for train data

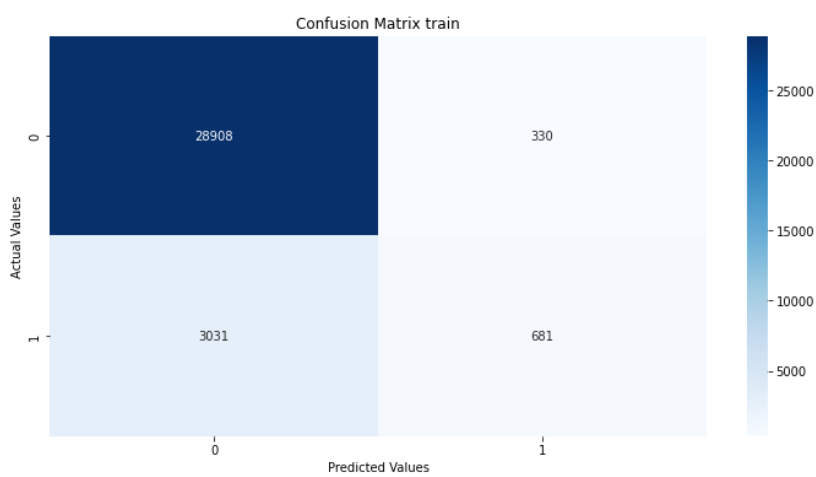


Figure 2.d.2 Confusion matrix for test data

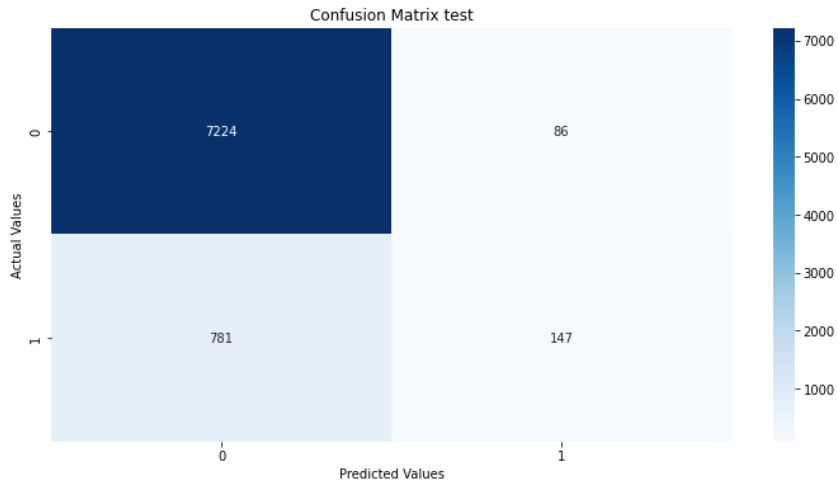


Table 2.d. Three Classification tables

(Optimal)	Actual		
Predicted	**Prob = 0.1168**	Yes	No
	Yes	2732	1908
	No	6171	30377

	Actual		
Predicted	**Prob = 0.3505**	Yes	No
	Yes	1417	3223
	No	1186	35362

	Actual		
Predicted	**Prob = 0.7504**	Yes	No
	Yes	289	4351
	No	87	36461

Table 3.b. Comparisons of Model Performances on Test Data

	Cutoff	Best Accuracy	Sensitivity	Specificity	AUC	Pseudo R^2
Group1-3	0.55	89.47%	15.84%	98.82%	0.72	0.1246
Group1-4	0.53	89.41%	15.30%	98.82%	0.77	0.1730

Table 3.c. Cross-tabulation Analysis Summary on Socio-economic data

	Chi-squared	P-value
emp_var_rate_group	2546.873	2.2e-16
cons_price_idx_group	1156.586	2.2e-16
cons_conf_idx_group	3037.675	2.2e-16
euribor3m_group	3494.457	2.2e-16
nr_employed_group	4530.276	2.2e-16

Figure 3.c. Cross-tabulation Analysis Output Result

3.1 Cross-tab: emp_var_rate_group

emp_var_rate_group	*y* no	*y* yes	***Total***
above0	22891 95.4 %	1106 4.6 %	23997 100 %
below0	13657 79.4 %	3534 20.6 %	17191 100 %
Total	36548 88.7 %	4640 11.3 %	41188 100 %

χ^2=2546.873 · df=1 · φ=0.249 · p=0.000

3.2 Cross-tab: cons_price_idx_group

cons_price_idx_group	*y* no	*y* yes	***Total***
93to94	24453 92.6 %	1961 7.4 %	26414 100 %
above94	4890 84.6 %	892 15.4 %	5782 100 %
below93	7205 80.1 %	1787 19.9 %	8992 100 %
Total	36548 88.7 %	4640 11.3 %	41188 100 %

χ^2=1156.586 · df=2 · Cramer's V=0.168 · p=0.000

3.3 Cross-tab: cons_conf_idx_group

cons_conf_idx_group	*y* no	*y* yes	***Total***
-35to-30	1082 56.5 %	833 43.5 %	1915 100 %
-40to-35	12905 92.7 %	1010 7.3 %	13915 100 %
-45to-40	14501 92 %	1267 8 %	15768 100 %
above-30	428 59.9 %	286 40.1 %	714 100 %
below-45	7632 86 %	1244 14 %	8876 100 %
Total	36548 88.7 %	4640 11.3 %	41188 100 %

χ^2=3037.675 · df=4 · Cramer's V=0.272 · p=0.000

3.4 Cross-tab: euribor3m_group

euribor3m_group	*y* no	*y* yes	***Total***
2to4	13 92.9 %	1 7.1 %	14 100 %
above4	26338 95.2 %	1338 4.8 %	27676 100 %
below2	10197 75.5 %	3301 24.5 %	13498 100 %
Total	36548 88.7 %	4640 11.3 %	41188 100 %

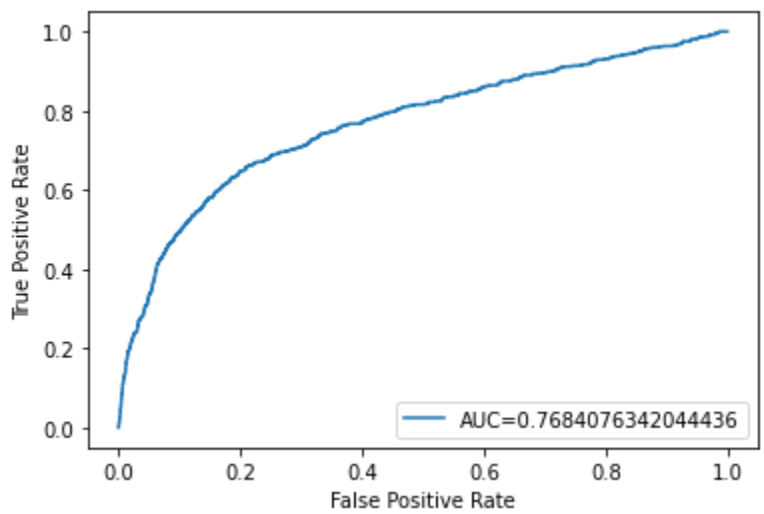
χ^2=3494.457 · df=2 · Cramer's V=0.291 · Fisher's p=0.000

3.5 Cross-tab: nr_employed_group

nr_employed_group	*y* no	*y* yes	***Total***
5000to5100	9493 78.5 %	2597 21.5 %	12090 100 %
5100to5200	10983 95.9 %	473 4.1 %	11456 100 %
above5200	15368 94.7 %	866 5.3 %	16234 100 %
below5000	704 50 %	704 50 %	1408 100 %
Total	36548 88.7 %	4640 11.3 %	41188 100 %

χ^2=4530.276 · df=3 · Cramer's V=0.332 · p=0.000

Figure 3.d. ROC Curve for test data Group (1-4)



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Final Page

Grade: _____